India's LPG Shortage Crisis 2026

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Abstract

This report investigates the LPG supply shortage that hit India in early 2026, driven by a global crude oil spike that pushed Brent prices to a peak of $118.35/bbl. Through a combination of web scraping, data cleaning, exploratory analysis, and machine learning, we analyze how crude oil prices and port distances jointly determine commercial LPG costs across Indian states. A Random Forest model is trained to predict prices and a Streamlit dashboard makes the findings interactive. The findings show that inland states face a compounding logistics penalty on top of the global fuel shock.

CCS Concepts

• **Mathematics of computing** → **Probability and statistics**; • **Computing methodologies** → *Machine learning*.

Keywords

LPG pricing, supply chain, Random Forest, web scraping, India energy crisis, price prediction, Streamlit

1 Data Collection & Web Scraping

1.1 Overview

Data for this project was collected from multiple online sources using Python-based web scraping. The goal was to build a unified dataset linking LPG prices, crude oil benchmarks, and logistics variables (port distances) across Indian states.

1.2 Primary Sources

The following websites were used as data sources:

- **My LPG** (`mylpg.in`) — State-wise commercial and domestic LPG cylinder prices for February, March, and April 2026.
- **Yahoo Finance / yfinance API** — Historical daily closing prices for Brent Crude Oil Futures (`BZ=F`) from October 2025 to April 2026.
- **Petroleum Planning & Analysis Cell (PPAC)** — National-level LPG consumption data (April 2025 to March 2026).
- **Manually compiled lookup table** — Distances from each state capital to its nearest major LPG import port (Mumbai, Kochi, Vizag, Paradip, Kandla, Kolkata).

1.3 Scraping Approach

Most data was scraped using the `BeautifulSoup` (`bs4`) library in Python. The standard workflow was: send an HTTP request using `requests`, parse the HTML tree with BeautifulSoup, locate price tables by CSS class or tag patterns, and extract relevant rows into a `pandas` DataFrame.

One website posed a challenge — the price table was rendered client-side via JavaScript, meaning a simple HTTP request returned a skeleton page without actual data. To work around this, the full rendered HTML was saved locally via the browser's "Save As" feature, and BeautifulSoup was run on that downloaded file. This is a practical workaround when a site blocks automated rendering but the data itself is not behind a login wall.

1.4 Final Dataset

After scraping, all datasets were merged into a single master file: `master_lpg_analysis_dataset_final.csv`. Each row represents one state in one month and contains: commercial LPG price, domestic LPG price, average Brent crude price for that month, and distance to the nearest import port in kilometres.

2 Data Cleaning

2.1 What Needed Cleaning

Raw scraped data is rarely ready for analysis. The key cleaning steps performed were:

- Price columns scraped as strings (e.g., `2,249.00`) — stripped of currency symbols and commas, then cast to `float`.
- State names had inconsistent formatting across source files (e.g., *Tamil Nadu* vs. *Tamilnadu*) — standardised to a common format for reliable merging.
- A few states had missing price entries for certain months — handled using forward-fill within each state group, appropriate given the slow rate of price change.
- Brent crude data had weekend/holiday gaps — resampled to monthly averages using `pandas resample()`, which smooths out daily noise.
- Port distance values were manually entered and cross-verified against Google Maps for three spot-checks.

2.2 Merge Strategy

The three separate scraped DataFrames (price data, crude data, port distance lookup) were merged using `pandas merge()` on the state name column. A left join on the price table preserved all state–month combinations and flagged states that failed to match — these were fixed by correcting the state name inconsistencies mentioned above.

The final cleaned dataset had no missing values in the core analytical columns (commercial price, domestic price, Brent crude, port distance) and covered 28 states across 3 months.

3 Exploratory Data Analysis & Visualizations

This section walks through the key visualizations generated during the EDA phase. Each plot was built to answer a specific analytical question about the crisis.

3.1 Brent Crude Oil — The Trigger

The first question was: what caused this spike? The Brent crude plot makes it immediately clear. Prices were stable in the $60–70/bbl range from October 2025 through February 2026, then shot up sharply from March, peaking at $118.35/bbl in April 2026. This

global supply shock is the root cause of everything downstream in this crisis.

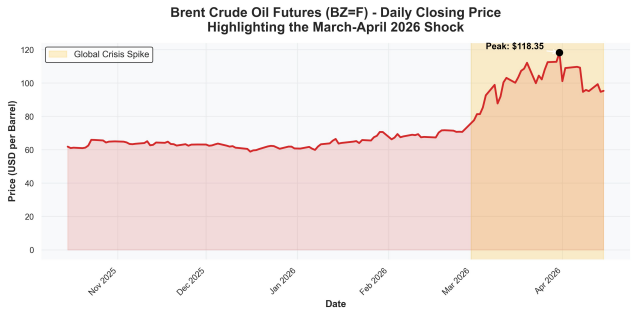

Figure 1: Brent Crude Oil Futures (BZ=F) daily closing prices — highlighting the March–April 2026 global crisis spike.

3.2 National LPG Consumption — The Warning Sign

Before prices went up, consumption went down. The consumption bar chart shows a sharp dip in March 2026 (2,379 KT) compared to the December–January peak (3,067 KT). This supply contraction likely reflects hoarding behaviour, distributor stock buildup ahead of expected price hikes, and early signs of shortage before the official price revision.

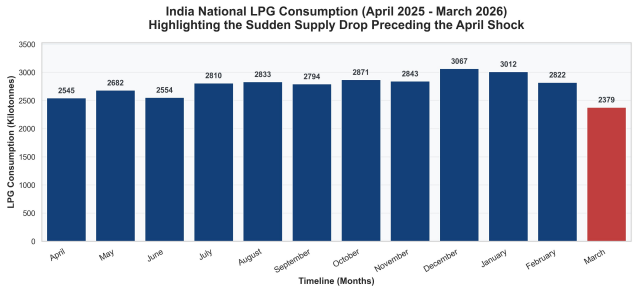

Figure 2: Monthly LPG consumption in India (April 2025 – March 2026). The red bar highlights the March 2026 supply drop preceding the price shock.

3.3 Price Trends — Commercial vs. Domestic

The price trend chart compares average commercial and domestic LPG prices from February to April 2026. Commercial prices rose steeply — from around 1,895 to 2,238 — while domestic prices stayed relatively flat (920 to 960). This gap reflects the government subsidy buffer applied to domestic cylinders, but the commercial sector absorbed the full brunt of the crude shock. Small restaurants, dhabas, and street vendors who use commercial cylinders faced an effective 18% cost jump in just two months.

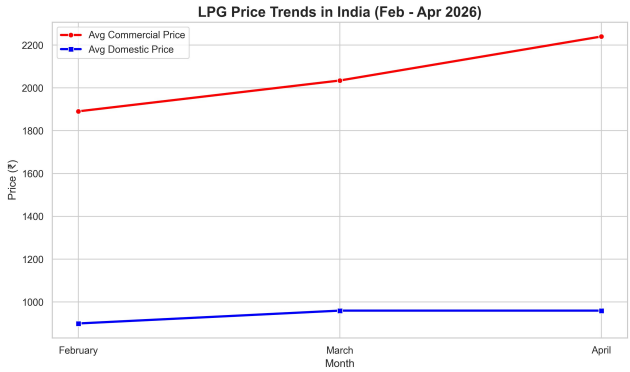

Figure 3: LPG price trends in India (Feb–Apr 2026). Commercial prices (red) surged while domestic prices (blue) remained partly shielded by government subsidies.

3.4 Geographic Distribution — The India Map

The India map plot visualises commercial LPG prices across states using a colour scale (yellow to dark red) with bubble sizes proportional to port distance. States in the northeast — Arunachal Pradesh, Nagaland, Mizoram — show the darkest colours (prices above 2,400) paired with large bubbles, confirming that geographic remoteness amplifies the base fuel price increase. Coastal and western states cluster in the yellow–orange range, benefiting from shorter supply chains.

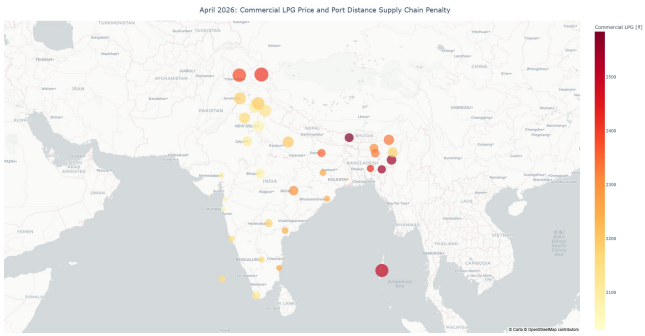

Figure 4: April 2026 commercial LPG prices across India. Bubble size represents distance to nearest port; darker colour indicates higher price.

3.5 Port Distance Effect — Scatter Plot

This scatter plot directly quantifies the logistics penalty. The positive slope of the regression line confirms that states further from import ports tend to pay more for commercial LPG. While the relationship is not perfectly linear (some inland states have regional buffer depots that moderate prices), the overall trend holds. States clustered above the regression band — high price despite moderate distance — are those with additional state-level taxes or limited distributor competition.

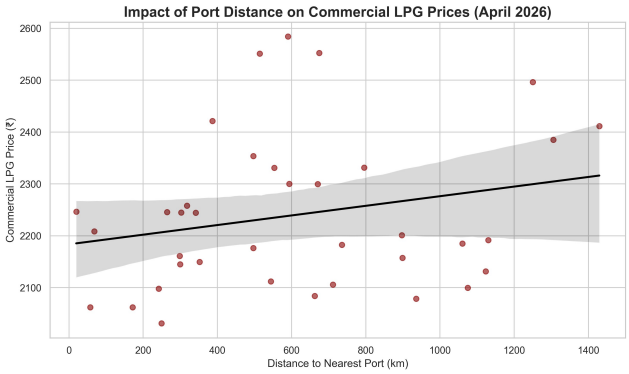

Figure 5: Commercial LPG price vs. distance to nearest import port (April 2026). The trend line shows a measurable positive logistics penalty.

4 Model Building & Dashboard

4.1 Problem Setup

The modelling question is straightforward: given a state's distance to the nearest LPG import port and the prevailing Brent crude oil price, can we predict what the commercial LPG price will be?

The two input features are:

- **Distance_to_Port_km** — captures the logistics cost embedded in the final cylinder price.
- **Avg_Brent_Crude_USD** — captures the global supply-side cost, which is the dominant driver during a price shock.

The target variable is *Commercial Price* (per 19 kg cylinder).

4.2 Why Random Forest?

A basic Linear Regression was tried first. It gave reasonable results but struggled with the non-linear interactions between crude price and distance — for example, distance matters much more when crude is already high. Random Forest handles this naturally.

A Random Forest builds 100 independent decision trees, each trained on a random subset of the data. The final prediction is the average across all 100 trees. This ensemble approach reduces overfitting and handles non-linear feature interactions without manual tuning.

4.3 Training & Evaluation

The dataset was split 80/20 into training and test sets using `train_test_split()` with a fixed random seed for reproducibility.

Table 1: Random Forest Model Evaluation Metrics

Metric	Score
R-squared (R^2)	≈ 0.62 (test set)
Mean Absolute Error	80–100 per cylinder
Cross-Validation Score (5-fold)	≈ 0.58

The R^2 of ≈ 0.62 means the model explains about 62% of price variance using just two features. Given that LPG prices are also influenced by state taxes, subsidy policies, and distributor markups not captured in the dataset, this is a reasonable result. The MAE of 80–100 means the model's price estimates are off by around 80–100 on average — acceptable for a macro-level planning tool.

4.4 Actual vs. Predicted Plot

Figure 6 compares the model's predictions to actual test prices. Points lying close to the diagonal (perfect prediction line) indicate good model fit. Most points cluster reasonably near the line, with some scatter — consistent with an R^2 of ≈ 0.62.

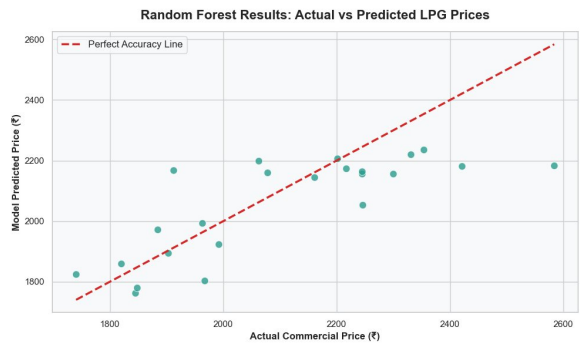

Figure 6: Random Forest model — Actual vs. Predicted commercial LPG prices. The dashed line represents perfect prediction.

4.5 Interactive Dashboard (Streamlit)

To make the model accessible beyond a static report, an interactive dashboard was built using Streamlit. The dashboard allows users to adjust two inputs via sidebar sliders: Brent Crude Price ($/bbl) and Delivery Distance (km). Based on these inputs, the model instantly predicts the commercial LPG price, classifies market status as *Stable*, *Elevated*, or *Critical*, and generates a live price sensitivity chart and a risk heatmap.

A "Market Shock" toggle is also included — setting crude to $140 and maximum distance simultaneously — to simulate worst-case disruption scenarios. The dashboard is built as a single `app.py` file, deployable via Streamlit Cloud or any Python server.

5 Conclusion

The 2026 LPG crisis was not a single-cause event. It was a global crude oil shock amplified through India's import-dependent supply chain, with geographic distance acting as a multiplier that hit inland and northeastern states hardest. The analysis establishes three key findings:

(1) The Brent crude spike from March 2026 is the primary driver of commercial LPG price increases across all states.
(2) Port distance creates a measurable logistics penalty, adding up to 200–300 per cylinder in the most remote states.
(3) The domestic subsidy shielded household consumers from the worst of the shock, but commercial users — including the informal food sector — bore the full cost.

Key Takeaway. In a crisis like this, data science is most useful not for perfect prediction but for rapid situational awareness —

understanding which states need urgent intervention, how much of the price rise is structural vs. logistical, and what policy levers are available. This project is a small proof-of-concept of that idea.

Project Repository

The complete project code is available at:
https://github.com/KanishkNandeshwar/Major-Project-LPG-shortage-2026#

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